Reduced Substation Models for Generalized State Estimation

Antonio Gómez Expósito and Antonio de la Villa Jaén

***Abstract*—This paper deals with the problem of including detailed substation models in state estimators so that topological errors and/or bad internal data can be detected and identified. Instead of analyzing constraints and state variables related to individual circuit breakers, the whole substation is reconsidered in the light of topological properties, leading to a model where zero-injection constraints are not explicitly handled. In addition, it is shown how, if the new state variables are selected keeping certain rules in mind, the resulting constraints can be easily transformed into equivalent constraints, many of which appear as critical. This allows a reduced model to be used without losing any capability to perform error analysis.**

***Index Terms*—State estimation, substation modeling, topological errors, topology processor.**

I. Introduction

INCORPORATION of circuit breaker (CB) statuses within the state estimation flow diagram, and related issues such as detection and identification of topology errors, is now a twenty-year-old topic [1], [2]. In these two decades, many contributions have been reported providing different and imaginative solutions to this problem.

Essentially, all methods proposed so far can be classified according to the model adopted. While some works try to extract the relevant information from a conventional bus-branch model [1], [3], [4]–[9] others perform detailed physical-level modeling of certain substations [2], [10]–[15]. Most usually, when the latter approach is adopted, the state estimation process comprises two steps, namely: 1) Conventional state estimation and bad data analysis. 2) State estimation including full models of suspicious substations. Some cases have been reported showing that state estimators are prone to convergence problems in the presence of certain topology errors. As a result, the above two-step procedure may not be viable, which has prompted several authors to develop alternative methods [4], [8] and a topology estimator based on an approximate power system model [15].

Systematic ways of modeling arbitrary combinations of switching devices, so as to be able to include in the model all existing measurements/constraints, are now well established [16]. A worth noting contribution in this regard refers to the role and significance of normalized Lagrange multipliers in the process of identifying topology errors [17]–[19]. It is also possible to check *a posteriori* the status of doubtful devices by means of hypothesis testing [20]. Recently, a production-grade state estimator has been described incorporating the most recent contributions in this field [21].

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The authors are with the Department of Electrical Engineering, University of Seville, Spain (e-mail: exposito@cica.es; avilla@esi.us.es).

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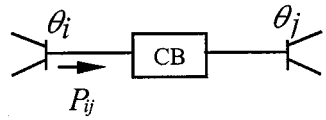

Fig. 1. Circuit breaker and related variables.

The goal of this paper is to reconsider the way individual CBs are modeled, in an attempt to obtain as reduced a substation model as possible. The next section briefly reviews existing CBs and substation models. Section III formally defines the substation graph, on which the proposed techniques are based. Sections IV and V deal with the minimum number of state variables and equality constraints that should be added when a substation is modeled at the physical level. Section VI explains how the new state variables should be chosen so that the substation model can be further reduced by omitting critical constraints. A test case is presented in Section VII showing the reduction achieved by applying the proposed techniques, and a numerical example is worked out in the Appendix.

II. Circuit Breaker Modeling

This section is intended to very briefly review the state of the art on CB modeling. The reader is referred to [10]–[12], [16], [21], [22] for further details. For simplicity, only the DC active subproblem will be discussed, although any noteworthy difference with the reactive model will be remarked.

Any CB can be explicitly modeled (Fig. 1) by adding its power flow, P_{ij}, to the state vector. Note also that two regular state variables (θ_i, θ_j) are involved. Compared to the output of a conventional Topology Processor, this model requires one extra state variable (P_{ij}) for an open device but two extra variables (θ_j, P_{ij}) for a closed one. The advantage is that any available measurement/constraint can be handled and subsequently checked if redundancy permits. The constraint $\theta_i - \theta_j = 0$ or $P_{ij} = 0$ is added to model closed or open status respectively. Furthermore, irrespective of the CB status, it is sometimes possible to add extra constraints. For instance, if node j in Fig. 1 is connected to a single branch, most likely it will be a zero-injection node $(P_j = 0)$.

In practice, a number of CBs and bus sections are interconnected within a substation, leading to one or several Bus Section Groups [21]. An example with five CBs along with the symbology adopted is shown in Fig. 2. In this case, the state vector contains the following 10 variables:

$$x = [\theta_1, \theta_2, \theta_3, \theta_4, \theta_5, P_{12}, P_{14}, P_{25}, P_{34}, P_{35}]$$

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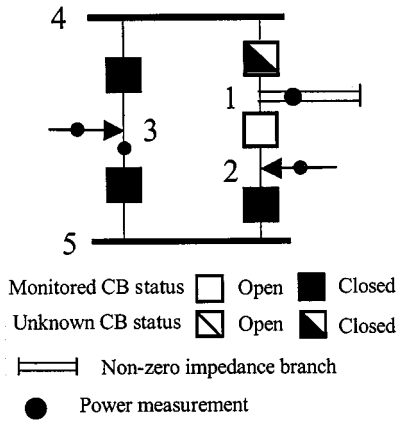

Fig. 2. Bus section group with 5 CBs and symbology adopted.

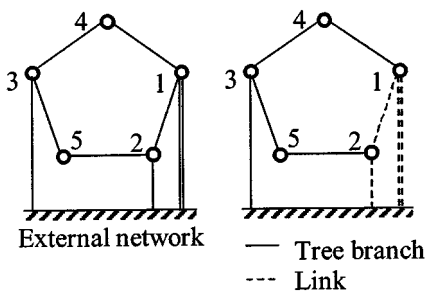

Fig. 3. Substation graph and power flow tree.

whereas a conventional state estimator would need a single state variable (θ_1) if CB 1–4 is assumed closed or two variables (θ_1, θ_2) if it is assumed open. To compensate for the extra variables, 7 equality constraints are enforced (3 of them corresponding to zero-injection nodes).

In general, for a group with s CBs and n nodes, a total of $s+n$ state variables would be required. The number of constraints depends on the number of zero-injection nodes and CBs whose status is known.

III. Substation Graph

In order to better convey the ideas presented below, the so-called **Substation Graph** (SG) will be introduced first (in this paper, the words "substation" and "bus section group" will be used indistinctly).

The SG is composed of $n+1$ vertices [2]: n bus sections plus a virtual node representing the external network. The n actual nodes are interconnected by s internal branches representing CBs. In addition to this internal subgraph, two types of external branches may exist: 1) n_i branches connecting nonzero injection nodes to the external node; 2) n_z nonzero impedance branches (regular branches). Collectively, both sets of branches will be referred to as the external subgraph. Fig. 3 shows the SG corresponding to the example of Fig. 2 ($n_i = 2, n_z = 1, s = 5$).

On this graph, many different trees can be chosen. By definition, any connected tree contains n tree branches and, consequently, determines $(s + n_i + n_z) - n$ links. As will be seen below, the structure and sparsity of the resulting equations depend not only on the available set of measurements and constraints, but on the selected tree as well.

IV. Minimally Augmented State Vector

Every branch of the SG contributes two unknowns to the active problem, namely its power flow and its voltage phase drop. This full set of unknowns must exactly satisfy all available constraints and, in the least-squares sense, all available measurements. Three types of constraints can be distinguished: 1) Linear constraints provided by CB statuses; 2) n_z nonlinear constraints coupling the power flow through every regular branch with the respective voltage phase difference; 3) Linear connectivity constraints (i.e., Kirchhoff laws). Notice that the two types of unknowns become coupled by the nonlinear constraints only. Depending on which constraints are explicitly retained, different models are obtained:

1) The standard bus/branch model results when all constraints are used to reduce the number of unknowns.
2) The bus section/switch model arises when only the first set is retained. The other two sets are of no interest in topological error analysis and can be eliminated.
3) Eventually, both parameter and topological errors could be tested if only Kirchhoff laws were employed to reduce the state vector.

The goal of this section is to analyze the resulting state vector when the second model is adopted.

Kirchhoff's voltage law allows any closed constraint $(\theta_{ij} = 0)$ or available voltage measurement to be expressed in terms of the n voltage differences of the branches of a tree. Such a tree will be termed the **Voltage Tree**. In practice, the n nodal phase angles, corresponding to a virtual star-shaped tree, serve for the same purpose. Therefore, no matter how many CBs exist, n phase angles are included in the state vector.

In a dual manner, Kirchhoff's current law allows any open constraint $(P_{ij} = 0)$ or available power flow measurement to be expressed solely in terms of the power flows of the links of a tree, that will be termed the **Power Flow Tree**. As explained in Section III, any connected tree of the SG leads to $(s + n_i + n_z) - n$ links. However, since power flows through regular branches are nonlinear functions of phase angles, already included in the state vector, the number of power flows strictly required as state variables can be reduced to $s + n_i - n$ by forcing these branches to be links and resorting to the respective nonlinear constraint (hence, regular branches should belong to the voltage tree but not to the power flow tree). Of these links, $n_i - 1$ are external injections and the remaining $s - n + 1$ correspond with CB power flows (internal links). The fact that impedances of internal branches are zero or undefined explains why internal links are required, in addition to external injections, to completely determine any power flow.

The total number of state variables is, therefore, limited to $s + n_i$, instead of the $s + n$ of existing formulations. Note that the reduction comes from eliminating the zero-injection constraints within the substation, which should be otherwise employed. This is clearly seen by writing $s+n_i-n$ as $s-(n-n_i)$, i.e., CBs minus zero-injection nodes. Unlike CB constraints, zero-injection constraints are never wrong and, therefore, no information is lost when they are eliminated.

A power flow tree for the previous example is also shown in Fig. 3 (the voltage tree is trivial when nodal phase angles are

used as state variables). With this particular tree, one power flow and one power injection are added to the state vector. It should be realized, however, that any other power flow tree could be chosen so long as nonzero finite impedances are excluded. For instance, if the selected tree contains the two injections, then two CB links appear as state variables. In both cases, three variables are saved with respect to current approaches.

Let z_e, θ_e denote the sets of external measurements and state variables respectively. The resulting DC model for the former example is composed of four equality constraints:

$$\begin{bmatrix} P_{12} \\ \theta_{25} \\ \theta_{35} \\ \theta_{34} \end{bmatrix} = \left[\begin{array}{cc|ccccc|c} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & 1 & -1 & 0 & 0 \end{array}\right] x = 0 \quad (1)$$

plus the available measurements

$$\begin{bmatrix} P_2 \\ P_3 \\ P_{35} \\ z_e \end{bmatrix} = \left[\begin{array}{cc|ccccc|c} 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & b_{1e} & 0 & 0 & 0 & 0 & * \\ -1 & -1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & * & 0 & 0 & 0 & 0 & * \end{array}\right] x + \varepsilon \quad (2)$$

where

$$x = [P_{12} \quad P_2 \quad | \quad \theta_1 \quad \theta_2 \quad \theta_3 \quad \theta_4 \quad \theta_5 \quad | \quad \theta_e]^t$$

the symbol $*$ denotes nonzero elements corresponding to external variables/equations (several rows/columns in the general case) and b_{1e} is the inverse of the line reactance.

The former development is based on the hypothesis that $n_i > 0$ (i.e., there is at least one injection connecting the internal tree to the external node). When $n_i = 0$, the power flow tree is disconnected from the external node, because regular branches are forced to be links. In such a case, as many power flows as CB links, $s - n + 1$, constitute state variables, and the state vector comprises a total of $s + 1$ elements. The extra variable with respect to the connected case is compensated by a conventional nonlinear equality constraint, forcing the sum of power flows leaving the substation to be zero. Alternatively, a zero-injection pseudomeasurement with high weight could be added to any internal node.

V. Linearly Independent Equality Constraints

Let

$$Ax = 0 \quad (3)$$

represent the set of linear constraints imposed by CB statuses. Irrespective of matrix A being of full rank or not, those constraints can be considered in the SE as very accurate measurements. The augmented objective function becomes

$$J(x) + \varrho x^t A^t A x \quad (4)$$

where $J(x)$ comprises the conventional measurements and ρ is much larger than regular weights. The state vector that minimizes the scalar function given by (4) very accurately satisfies (3).

Alternatively, a Lagrangian explicitly considering (3) can be minimized

$$J(x) - \lambda^t \cdot Ax. \quad (5)$$

Comparing (4) and (5) it can be concluded that

$$\lambda = -\varrho \cdot (Ax). \quad (6)$$

As ρ tends to infinity, the norm of the residual vector (Ax) tends to zero, but their product approaches the vector of Lagrange multipliers [17], [18].

The first-order optimality conditions, when applied to (5), allow both the state vector, x, and Lagrange multipliers, λ, to be computed, provided all equality constraints are linearly independent (matrix A of full rank).

Assume now that a set of redundant constraints

$$Bx = 0 \quad (7)$$

is considered, in addition to (3). The Lagrangian becomes

$$J(x) - \lambda_A^t \cdot Ax - \lambda_B^t \cdot Bx \quad (8)$$

but, since the rows of B are linear combinations of the rows of A,

$$B = k \cdot A \quad (9)$$

(8) can be also written as

$$J(x) - (\lambda_A + k^t \lambda_B)^t Ax. \quad (10)$$

This means that

$$\lambda = \lambda_A + k^t \lambda_B \quad (11)$$

or, in other words, that only the Lagrange multipliers of linearly independent constraints can be computed. The individual components λ_A and λ_B can be obtained, however, if additional assumptions are considered. For instance, modeling all available constraints as equally weighted virtual measurements

$$J(x) + \varrho x^t A^t A x + \varrho x^t B^t B x \quad (12)$$

translates into the following relationship

$$\lambda_B = k \lambda_A. \quad (13)$$

Substituting (13) into (11) yields

$$(I + k^t k)\lambda_A = \lambda. \quad (14)$$

Therefore, when the objective function (5), rather than (4), is minimized, only linearly independent constraints should be used. The Lagrange multipliers corresponding to arbitrary sets of constraints could be obtained, if needed, by means of (14) and (13).

Much in the same way and for the same motivation as in the state-variable approach to circuit analysis, the so-called **Proper Tree** is introduced at this point. It provides an automatic procedure for selecting the largest set of linearly independent equality

constraints. The proper tree is built keeping in mind the following rules:

- Include as many closed CBs as possible
- Exclude as many open CBs as possible.

The power flow tree of Fig. 3 has been chosen in such a way that it is also a proper tree.

When the proper tree contains, as in this case, all closed but none open CB, all equality constraints are linearly independent. Otherwise, there are redundant constraints that should be excluded from the model (loops of closed CBs and cut-sets of open CBs).

In the sequel, it will be assumed that only linearly independent equality constraints, identified by means of the proper tree, are retained.

VI. Equivalent Constraints and Reduced Models

Examining (1) and (2), it is evident that the two constraints $\theta_{34} = 0$ and $\theta_{25} = 0$ are critical to make θ_4 and θ_2 observable. This means that the associated multipliers are null and, therefore, both the constraints and the critically observable variables can be removed from the model beforehand.

In the general case, this idea can be extended to sets of critical constraints and critically observable state variables. Assume the CB constraints can be written as

$$\begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} A_1 & A_2 \\ 0 & B_2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0 \tag{15}$$

and the remaining nonlinear constraints and available measurements, when linearized, take the form

$$z = [0 \quad H] \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \varepsilon \tag{16}$$

where A_1 is a square matrix of full rank. When this happens, the set of constraints c_1 is critical to estimate x_1 and the following reduced model can be adopted

$$c_2 = B_2 x_2 = 0 \tag{17}$$

$$z = H x_2 + \varepsilon. \tag{18}$$

Notice that there is no way to detect wrong statuses for CBs belonging to the set c_1 ($\lambda_1 = 0$).

It is not frequent, however, for the CB constraints to appear as in (15), the following structure being more common

$$\begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} A_1 & A_2 \\ B_1 & B_2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0. \tag{19}$$

However, so long as A_1 is a regular matrix, the original constraints (19) can be transformed into equivalent constraints with the desired form by means of the following transformation matrix

$$T = \begin{bmatrix} I & 0 \\ -B_1 A_1^{-1} & I \end{bmatrix} \tag{20}$$

yielding,

$$\begin{bmatrix} c_1' \\ c_2' \end{bmatrix} = T \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} A_1 & A_2 \\ 0 & B_2' \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0 \tag{21}$$

where $B_2' = B_2 - B_1 A_1^{-1} A_2$. The reduced model (17), (18) remains valid substituting B_2 by B_2'. An interesting particular case arises when $A_2 = 0$, because $B_2' = B_2$.

Once the reduced model is solved, the Lagrange multipliers of the actual constraints can be obtained from

$$\begin{bmatrix} \lambda_1 \\ \lambda_2 \end{bmatrix} = T^t \begin{bmatrix} \lambda_1' \\ \lambda_2' \end{bmatrix}$$

and the covariance matrix

$$\mathrm{cov} \begin{bmatrix} \lambda_1 \\ \lambda_2 \end{bmatrix} = T^t \, \mathrm{cov} \begin{bmatrix} \lambda_1' \\ \lambda_2' \end{bmatrix} T.$$

Furthermore, taking into account that $\lambda_1' = 0$ and that the associated covariance submatrix is null, i.e.,

$$\mathrm{cov} \begin{bmatrix} \lambda_1' \\ \lambda_2' \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & \mathrm{cov}_{22} \end{bmatrix}$$

yields

$$\begin{aligned} \lambda_1 &= -\left(B_1 A_1^{-1}\right)^t \lambda_2' \\ \lambda_2 &= \lambda_2' \end{aligned} \tag{22}$$

and

$$\mathrm{cov} \begin{bmatrix} \lambda_1 \\ \lambda_2 \end{bmatrix} = \begin{bmatrix} \left(B_1 A_1^{-1}\right)^t \mathrm{cov}_{22} B_1 A_1^{-1} & -\left(B_1 A_1^{-1}\right)^t \mathrm{cov}_{22} \\ -\mathrm{cov}_{22} B_1 A_1^{-1} & \mathrm{cov}_{22} \end{bmatrix}. \tag{23}$$

The rest of this section will be devoted to discuss how the state variables can be systematically selected so that advantage can be taken of the reduced model. The key idea consists of choosing as state variables the measured quantities, in an attempt to get as sparse a Jacobian as possible (sparsity of the constraint coefficient matrices is not crucial as they can always be transformed). Whenever possible, the set of state variables should be completed with power flows and voltage phase drops of open and closed CBs, respectively.

Consequently, the voltage and power flow trees defined in Section IV should be chosen keeping in mind the following rules:

- Voltage Tree:
  - — Include all regular (nonzero impedance) branches. This way, the respective nodal phase angles, which are necessary to mathematically represent external measurements, are included in the state vector.
  - — In the reactive subproblem, every voltage measured node requires an extra branch to "ground" which should be included in the tree as well.
  - — Include as many closed CBs as possible (when all nodes are measured, or connected to regular branches, none CB can be included).
  - — Complete, if necessary, with unknown (or open) CBs or injections. From the point of view of Kirchhoff's voltage law, these branches of the SG are irrelevant and could be ignored (open-circuited).
- Power Flow Co-tree:
  - — Include all regular branches, measured or not. As discussed in Section IV, these branches do not contribute power flow variables, because they are functions of phase angles which do belong to the state vector.
  - — Include as many measured powers (flows or injections) as possible.

— Include as many open CBs as possible.
— Complete, if necessary, with unmeasured or unconstrained branches. From the point of view of Kirchhoff current law, these branches are irrelevant and could be short-circuited.

When these rules are followed, the resulting model has the following structure

$$\begin{bmatrix} c_{ct} \\ c_{ol} \\ c_{cl} \\ c_{ot} \end{bmatrix} = \left[\begin{array}{cc|cc|c} I & 0 & 0 & 0 & 0 \\ 0 & I & 0 & 0 & 0 \\ \hline C_1 & 0 & C_2 & 0 & 0 \\ 0 & D_1 & H_{ot} & D_2 & H_{oe} \end{array}\right] \left[\begin{array}{c} \theta_c \\ P_o \\ \hline \theta_T \\ P_L \\ \hline \theta_e \end{array}\right] = 0 \quad (24)$$

$$z = [0 \quad 0 \mid H_{zt} \quad D_L \mid H_{ze}] \left[\begin{array}{c} \theta_c \\ P_o \\ \hline \theta_T \\ P_L \\ \hline \theta_e \end{array}\right] + \varepsilon \quad (25)$$

where the symbology adopted is,

c_{ct}: Closed CBs included in the voltage tree.
c_{ol}: Open CBs included in the power flow co-tree.
c_{cl}: Closed CBs excluded from the voltage tree.
c_{ot}: Open CBs excluded from the power flow co-tree.
P_o: CB power flows belonging to the state vector.
P_L: Other power flow state variables.
θ_c: CB voltage phase drops belonging to the state vector.
θ_T: Other voltage phase state variables.
θ_e: External state vector components.
C_i, D_i: Topological submatrices directly obtained from the selected trees.
H_{ij}: Jacobian submatrix obtained from linearization of regular branch power flows.

Equation (24) constitutes a particular case of (19) with $A_2 = 0$ and $A_1 = I$. Therefore, by previous application of the transformation matrix

$$T = \begin{bmatrix} I & 0 & 0 & 0 \\ 0 & I & 0 & 0 \\ -C_1 & 0 & I & 0 \\ 0 & -D_1 & 0 & I \end{bmatrix} \quad (26)$$

the full substation model can be substituted by the equivalent reduced model

$$\begin{bmatrix} c'_{cl} \\ c'_{ot} \end{bmatrix} = \begin{bmatrix} C_2 & 0 & 0 \\ H_{ot} & D_2 & H_{oe} \end{bmatrix} \begin{bmatrix} \theta_T \\ P_L \\ \theta_e \end{bmatrix} = 0 \quad (27)$$

$$z = [H_{zt} \quad D_L \quad H_{ze}] \begin{bmatrix} \theta_T \\ P_L \\ \theta_e \end{bmatrix} + \varepsilon \quad (28)$$

where closed CBs belonging to the voltage tree and open CBs excluded from the power flow tree are missing.

The information pertaining to the original model, necessary to perform topological error analysis, can be obtained from the reduced model by means of (22) and (23). Note that, since $A_1 = I$, the computational effort of this step is very small.

As an example, consider the substation shown in Fig. 4, whose relevant trees are represented in Fig. 5. The CB

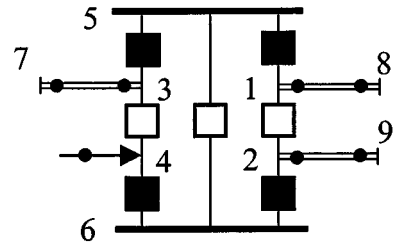

Fig. 4. 7-CB substation.

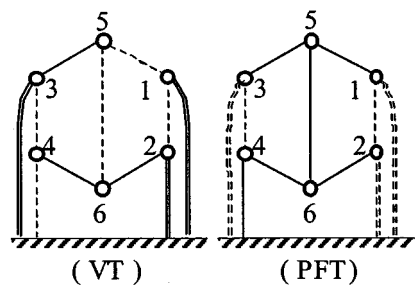

Fig. 5. Voltage and power flow trees for the 7-CB example.

TABLE I
NUMBER OF STATE VARIABLES AND CONSTRAINTS REQUIRED BY DIFFERENT MODELS

	B/b	Reduced	Full
Power flows	-	-	7
Phase angles	2	3	6
• Total Variables	2	3	13
Closed CB	-	1	4
Open CB	-	1	3
Zero-injection	1	-	5
• Total Constraints	1	2	12

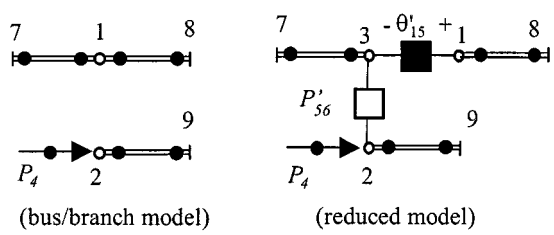

Fig. 6. Bus/branch and reduced model for the 7-CB example.

constraints, all of which are linearly independent in this case, and the set of measurements can be written as in (24), (25),

$$\begin{bmatrix} \theta_{35} \\ \theta_{46} \\ \theta_{26} \\ P_{12} \\ P_{34} \\ \theta_{15} \\ P_{56} \end{bmatrix} = \left[\begin{array}{ccccc|cccc} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ \hline 1 & 0 & 0 & 0 & 0 & 1 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 & -1 & -b_{1e} & 0 & -b_{3e} & * \end{array}\right] x = 0 \quad (29)$$

$$\begin{bmatrix} P_4 \\ z_e \end{bmatrix} = \left[\begin{array}{ccccc|cccc} 0 & 0 & 0 & 0 & 0 & b_{1e} & b_{2e} & b_{3e} & * \\ 0 & 0 & 0 & 0 & 0 & * & * & * & * \end{array}\right] x + \varepsilon \quad (30)$$

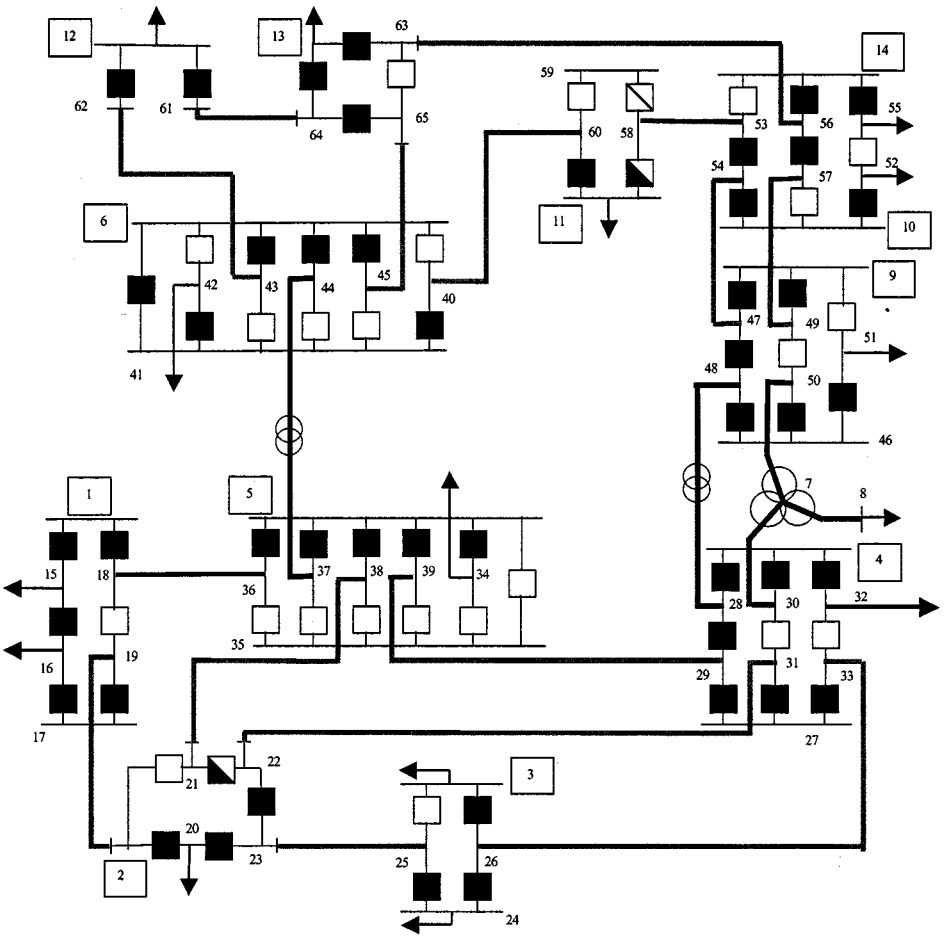

Fig. 7. Expanded 14-bus IEEE test system.

where

$$x = [\theta_{35} \quad \theta_{46} \quad \theta_{26} \quad P_{12} \quad P_{34} \mid \theta_1 \quad \theta_2 \quad \theta_3 \quad \theta_e]^t.$$

Hence, the reduced model becomes

$$\begin{bmatrix} \theta'_{15} \\ P'_{56} \end{bmatrix} = \begin{bmatrix} 1 & 0 & -1 & 0 \\ -b_{1e} & 0 & -b_{3e} & * \end{bmatrix} \begin{bmatrix} \theta_1 \\ \theta_2 \\ \theta_3 \\ \theta_e \end{bmatrix} = 0 \qquad (31)$$

$$\begin{bmatrix} P_4 \\ z_e \end{bmatrix} = \begin{bmatrix} b_{1e} & b_{2e} & b_{3e} & * \\ * & * & * & * \end{bmatrix} \begin{bmatrix} \theta_1 \\ \theta_2 \\ \theta_3 \\ \theta_e \end{bmatrix} + \varepsilon. \qquad (32)$$

Table I summarizes the number of state variables and constraints involved in the bus/branch model, the reduced model obtained above and the full model employed so far. The reduced model saves 10 variables/constraints compared to existing formulations, but retains the capability to detect and identify topology errors, whenever possible.

Fig. 6 shows, for comparison, the bus/branch substation model and an equivalent representation of the reduced model developed in this paper.

Of course, the larger the number of measurements, the higher the number of CB variables/constraints retained in the reduced model. For instance, the reader may check that the open CB 3–4 should appear in the model if P_{35} was measured. It should be noticed that, depending on the external configuration, the reduced model may still contain critical constraints. For example, $\theta'_{15} = 0$ in the previous example will be critical if the external network is radial, but not necessarily in the meshed case.

VII. Test Cases

The above ideas have been applied to the IEEE 14-bus test system. Every electrical bus has been expanded and modeled as a hypothetical but realistic substation. All typical arrangements have been employed in the model: ring, one-and-a-half CB, two bus sections with coupler, etc. (see Fig. 7).

Table II contains, for every substation, the relevant data concerning its topology and CB statuses. From left to right, the number of bus sections, injections, regular branches and CBs whose status is assumed open, closed or unknown, is shown.

Table III summarizes the number of state variables required in the reduced model for three measurement sets: 1) All injections but none CB power flow measured. 2) The indicated power flow measurement is added to set #1. 3) All closed CB power flows are measured, in addition to set #1. For comparison, the number of state variables required by the conventional full model is also included. In all cases, ordinary external branches are measured at both ends.

Table IV reports on the number of equality constraints which are handled by each model. Note that as many constraints as state variables are saved in all cases.

TABLE II
SUBSTATION TOPOLOGY DATA

Subst.	Bus Sec.	Injec.	Reg. br.	O	C	U
1	6	2	2	1	5	0
2	5	1	4	1	4	1
3	4	2	2	1	3	0
4	8	1	5	2	7	0
5	7	1	4	6	5	0
6	7	1	4	5	6	0
9	7	1	4	2	6	0
10/14	8	2	4	3	6	0
11	4	1	2	2	2	2
12	3	1	2	0	2	0
13	4	1	3	1	3	0

TABLE III
MEASUREMENT SET AND REQUIRED STATE VARIABLES

Subst.	Set #1	Set #2 CB meas.	Set #3	STATE VARIABLES			
				Full Mod.	Reduced Model		
					S#1	S#2	S#3
1	All Injections measured	1-18	Flow measured in all Closed CB	12	3	4	4
2		20-23		10	4	5	5
3		24-25		8	3	4	4
4		29-27		17	5	6	7
5		5-37		17	4	5	9
6		6-44		18	4	5	9
9		51-46		15	4	5	6
10/14		52-10		17	5	6	7
11		11-60		7	2	3	3
12		12-62		5	2	2	2
13		13-64		8	3	4	4
TOTAL				134	39	49	60

TABLE IV
MEASUREMENT SET AND REQUIRED CONSTRAINTS

	Full Model	Reduced Model		
		Set #1	Set #2	Set #3
Zero Injection	49	0	0	0
Closed CB	47	22	22	22
Open CB	22	1	11	22
TOTAL	118	23	33	44

As expected, the reduction achieved depends on measurement redundancy. As far as the state vector is concerned the reduction in size ranges from 71% for set #1 to 55% for set #3, which is the worst possible case. In practice, it is very unlikely that so many CB power flows are measured as in case #3.

All numerical experiments conducted confirm that the normalized residuals and Lagrange multipliers obtained with the reduced model are the same as those provided by the full model (see the Appendix).

TABLE V
NORMALIZED RESIDUALS AND MULTIPLIERS

Measurement or Constraint	Norm. Residual / Norm. Multiplier	
	Case # 1	Case # 2
P_4	12.9	6.5
$P_{12}=0$	20.4	4.1
$P_{34}=0$	20.4	4.1
$P_{56}=0$	20.4	4.1

VIII. CONCLUSION

In this paper, the incorporation of detailed substation models in state estimators is studied. Unlike earlier approaches, based on the inclusion of state variables and constraints contributed by every individual CB, an analysis is made of the substation as a whole, in terms of Kirchhoff's laws, leading to a minimally augmented state vector where zero-injection constraints are implicitly considered.

In addition, it is shown how the resulting model can be further reduced by using a set of equivalent constraints where many of them are critical and can be removed. The alternative set can be simply obtained from the original constraints if the state vector is augmented with strategically chosen variables.

Results are provided showing that over 50% reduction in the model size can be obtained in the worst case.

APPENDIX

This appendix presents some simulation results corresponding to the substation and measurement set of Fig. 4. In case #1, all measurements are exact but CB 5–6, modeled as open, is actually closed (split error). The topological constraints of the reduced model are given by (31). As $\theta'_{15} = 0$ is critical, its multiplier and respective row/column of the covariance matrix are null. On the other hand the multiplier of the second constraint $(P'_{56} = 0)$ is 833.3 and the corresponding diagonal of the covariance matrix is 1666.7. The transformation matrix, equation (26), is defined by the submatrices

$$C_1 = [1 \quad 0 \quad 0] \qquad D_1 = [-1 \quad -1].$$

With these values, the multipliers and covariances of the actual constraints are computed from (22), (23), and the results obtained are exactly the same as those of the full model. Table V shows the largest normalized residual for this case, corresponding to P_4, and the normalized multipliers corresponding to the 3 CBs assumed open. Note that the topological error is properly detected but, as no CB power flows are measured, it is not possible to identify which of the 3 CBs is closed. This explains why the 3 multipliers have the same value.

In case #2, there is no topological error, but P_4 is 50% higher than its true value. Repeating the process, the correct normalized residuals and multipliers are again obtained from the reduced model, showing the presence of bad data.

REFERENCES

[1] R. L. Lugtu, D. F. Hackett, K. C. Liu, and D. D. Might, "Power system state estimation: Detection of topological errors," *IEEE Trans. Power Apparatus and Systems*, vol. PAS-99, no. 6, pp. 2406–2412, Nov./Dec. 1980.

[2] M. R. Irving and M. J. H. Sterling, "Substation data validation," *IEE Proceedings*, pt. C, vol. 129, no. 3, pp. 119–122, May 1982.

[3] P. Bonanorni and G. Gramberg, "Power system data validation and state calculation by network search techniques," *IEEE Trans. Power Apparatus and Systems*, vol. PAS-102, no. 1, pp. 238–249, Jan. 1983.

[4] N. Singh and H. Glavitsch, "Detection and identification of topological errors in online power system analysis," *IEEE Trans. Power System*, vol. 6, no. 1, pp. 324–331, Feb. 1991.

[5] H.-J. Koglin and H.-T. Neisius, "A topology processor based on state estimation," in *Proceedings of the 11th Power Systems Computation Conference*, Avignon, Aug.–Sept. 1993, pp. 633–638.

[6] A. Simoes Costa and J. A. Leao, "Identification of topology errors in power system state estimation," *IEEE Trans. Power Systems*, vol. 8, no. 4, pp. 1531–1538, Nov. 1993.

[7] H. Singh and F. L. Alvarado, "Network topology determination using least absolute value state estimation," *IEEE Trans. Power System*, vol. 10, no. 3, pp. 1159–1165, Aug. 1995.

[8] J. C. S. Souza, A. M. Leite da Silva, and A. P. Alves da Silva, "On line topology determination and bad data suppression in power system operation using artificial neural networks," *IEEE Trans. Power System*, vol. 13, no. 3, pp. 796–803, Aug. 1998.

[9] A. G. Expósito, P. Z. Periñán, and V. H. Quintana, "Accurate estimation of circuit breaker positions," in *Proc. Large Engineering Systems Conference on Power Engineering*, Halifax, June 1998, pp. 11–14.

[10] A. Monticelli and A. Garcia, "Modeling zero impedance branches in power systems state estimation," *IEEE Trans. Power System*, vol. 6, no. 4, pp. 1561–1570, Nov. 1991.

[11] A. Monticelli, "The impact of modeling short circuit branches in state estimation," *IEEE Trans. Power System*, vol. 8, no. 1, pp. 364–370, Feb. 1993.

[12] ——, "Modeling circuit breakers in weighted least squares state estimation," *IEEE Trans. Power System*, vol. 8, no. 3, pp. 1143–1149, Aug. 1993.

[13] A. Abur, H. Kim, and M. K. Celik, "Identifying the unknown circuit breaker statuses in power networks," *IEEE Trans. Power System*, vol. 10, no. 4, pp. 2029–2037, Nov. 1995.

[14] Slutsker and S. Mokhtari, "Comprehensive estimation in power systems: State, topology, and parameter estimation," in *Proceedings of the American Power Conference*, vol. 57-I, Chicago, IL, Apr. 1995, pp. 149–155.

[15] L. Mili, G. Steeno, F. Dobraca, and D. French, "A robust estimation method for topology error identification," *IEEE Trans. Power System*, vol. 14, no. 4, pp. 1469–1476, Nov. 1999.

[16] A. Monticelli, *State Estimation in Electric Power System. A Generalized Approach*: Kluwer Academic Publishers, 1999.

[17] R. A. M. van Amerongen, "On the exact incorporation of virtual measurements in orthogonal-transformation base state-estimation procedures," *International Journal of Electrical Power & Energy Systems*, vol. 13, no. 3, pp. 167–174, June 1991.

[18] A. Gjelsvik, "The significance of the Lagrange multipliers in WLS state estimation with equality constraints," in *Proceedings of the 11th Power Systems Computation Conference*, Avignon, Aug.–Sept. 1993, pp. 619–625.

[19] K. A. Clements and A. S. Costa, "Topology error identification using normalized Lagrange multipliers," *IEEE Trans. Power Systems*, vol. 13, no. 2, pp. 347–353, May 1998.

[20] A. Monticelli, "Testing equality constraint hypothesis in weighted least squares state estimators," in *Proc. PICA-Conference*, Santa Clara, 1999, TS01_2.

[21] O. Alsac, N. Vempati, B. Stott, and A. Monticelli, "Generalized state estimation," *IEEE Trans. Power System*, vol. 13, no. 3, pp. 1069–1075, Aug. 1998.

[22] A. Villa and A. Gómez, "Modeling unknown circuit breakers in generalized state estimators," in Power Tech 2001 Oporto, Portugal, Sept. 2001, Paper accepted no. 221.

Antonio Gómez Expósito was born in Spain in 1957. He received the electrical engineering degrees from the University of Seville. Since 1982, he has been with the Department of Electrical Engineering, University of Seville, where he is currently a Professor. His primary areas of interest are sparse matrices, load flow, optimization, state estimation and computer relaying.

Antonio de la Villa Jaén was born in Spain in 1960. He received the electrical engineering degree from the University of Seville where he is pursuing the Ph.D. degree at the Department of Electrical Engineering. He has been a High-School teacher and he is presently a lecturer at the University of Seville. His primary areas of interest are computer methods for power system state estimation problems.